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St. Joseph's College of Arts & Science (Autonomous)

St. Joseph's College Road, Cuddalore – 607001

CH616 - PHYSICAL CHEMISTRY – III

Time : 3 hrs **Max Marks :75**

SECTION A (25x1=25)

Answer **ALL** questions:

I choose the correct answer **(10x1=10)**

1. For a strong electrolyte, the degree of dissociation is
   a) 1 b) 0 c) ∝ d) 0.5

2. The sum of the transport number of cation and axiom is equal to
   a) 1 b) 0 c) 0.5 d) ∝

3. The time required to pass 36,000 through an electro plating bath using a current of 5 A is

   a) 7.2 h b) 2 h c) 7.2 minutes d) 1 hr

4. A salt of weak acid and strong base on hydrolysis yielding a solution
   a) Slightly acidic b) Slightly c) Neutral d) Highly acidic

5. The P_H value of $Alcl_3$ is
   a) < 7 b) 77 c) 7 d) 0

6. Kohlrasch's law can be expressed as
   a) $\lambda\alpha = \lambda a - \lambda c$ b) $\lambda\alpha = \lambda c - \lambda a$
   c) $\lambda\alpha = \lambda\alpha + \lambda c$ d) $\lambda\alpha = \lambda a \times \lambda c$

7. For a spontaneous cell reation the emf of a cell should be
   a) +ve b) -ve c) Zero d) ∝

8. For a reaction that obey eienstein's law
   a) ∅=1 b) ∅>1 c) ∅<1 d) ∅=∝

9. The Henderson's equation used to calculate the p^{OH} of the basic buffer is

a. $P^{OH} = P^{kb} + \log\frac{[salt]}{[base]}$ b. $P^{OH} = P^{kb} - \log\frac{[salt]}{[base]}$

c. $P^{OH} = P^{kb} + \log\frac{[base]}{[salt]}$ d. $P^{OH} = P^{kb} - \log\frac{[base]}{[salt]}$

10. For phase and one component the degree of
    a) 1 b) 2 c) 3 d) 4

II Fill in the blanks **(5x1 = 5)**

11. The ionic product of water is _____

12. Nernst equation is _____

13. Phase rule is _____

14. Paramagretism is due to _____

15. Energy associated with radiation of wave length 1000^0C is _____

III Match the following **(5x1=5)**

16. Kohlrasch's law	a) $-\log[H^+]$
17. P_H	b) Ohm^{-1}
18. Phase rule	c) mg
19. Conductance	d) C-P+2=F

20. Photosensitizer c) $\lambda_+ + \lambda_-$

SECTION – B (10x2=20)

Answer any **FIVE** Questions

Choose the correct answer and justify it.

21. What is transport number?
22. State kohlraschcs law.
23. What is a buffer solution?
24. What is a Lewis acid?
25. What is a fuel cell?
26. What is LJP?
27. Define polarization.
28. State Grothes Draper's law.
29. What is fluorescence?
30. State phase rule.
31. Define quant ion yield.
32. What is the order of a reaction?

SECTION – C (5x6=30)

Answer **ALL** Questions

33. Explain Debye –Huckel Onsager theory of strong elecholytes.
34. Explain conductonetric titration between mixture of acids and strong base.

35. How does solubility product of a sparingly soluble salt is determined?
36. Explain the degree of hydrolysis of a salt of weal acid and strong bax.
37. Explain the kinetics of H_2 and B_{r2} reaction
38. Derive the emf of a concentration well with transference.
39. Explain the principle of Laser.

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